Composed Contraction Bounds and Quadratic-Time Cauchy Tables

A review of fifteen ProveIt ETH proposals,
with stronger resource certificates and a measured kernel improvement

Research prepared for Vladimir Reshetnikov
with ChatGPT

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Abstract

We review fifteen research packets from the requested directories: seven first-round packets and eight second-round packets, including the matching refinement added during this review. The final snapshot is `474ddba696a5`, after placement commit `6d3adc450615`. The strongest established logarithmic order in this corpus remains the density-sensitive bound $\mathrm{poly}_k(n,m)2^{O_k((n/\sqrt{\log n})\min\{\log(2+m/n),\log\log n\})}$; stronger denominator-$\log n$ proposals retain missing hypotheses.

We give three concrete improvements. First, composing asymmetric packet balancing, full crossing-block Cauchy modes, and an explicit multistage schedule gives a polynomial-space normalized contraction bound with exponent coefficient $63.857409\ldots$; the eventual coefficient 64 is certified by rational inequalities. The strongest individual generic polynomial-space certificate in the reviewed packets is $76.908995\ldots$. Second, a fully costed eight-group cached composition gives time coefficient $38.484496\ldots$ and space coefficient $34.274803\ldots$, on the same $s/\sqrt{\log_2 s}$ scale. Third, an incremental valuation/unit algorithm constructs the two full Cauchy feature tables in quadratic rather than cubic ring-operation work, with one modular unit inversion. Exact checks agree with the historical implementation, and reproducible local measurements show a 26.17-fold speedup against a matched modular baseline at dimension 128.

The contraction results retain the preprint's unchanged algebraic framework and the reviewed packing lemmas. They are ordinary mathematical derivations, not a new formal verification. The measured improvement is a kernel result, not an end-to-end SAT benchmark. No smaller unrestricted logarithmic order is established here.

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1 Results, scope, and comparison standard

The purpose of this iteration is to improve the strongest relevant frontier in both requested directories. The initial project README summarized seven packets. During the review, eight second-round packets were placed; their full proofs and code required a separate comparison with the earlier frontier. The current report supplies explicit resource bounds, executable code, exact arithmetic checks, historical-source provenance, and timings.

Write s for the normalized Boolean tensor-wire budget, n for SAT variables, and m for input clauses. Put $\lambda = \lceil \log_2 s \rceil$. The coefficients below multiply $s/\sqrt{\log_2 s}$ in a base-two exponent. They are *not* universal numerical coefficients on SAT variables. The reduction has its own width-dependent size constants.

Generic normalized contraction certificate	Time coefficient	Space
Strongest individual incoming polynomial-space packet	76.908995	polynomial
Fixed-$r = 16$ slope-four/asymmetric boundary composition	66.192724	polynomial
This report: explicit multistage schedule	63.857410	polynomial
This report: eight-group cached schedule	38.484496	34.274803

The second row is an optimized comparison within a specified fixed-arity schedule model; it is an infimum, not a lower bound on all algorithms. The last row deliberately uses subexponential space. Safe eventual coefficients are 64 for the polynomial-space method, and 39 time with 35 space for the cached method. The same polynomial-space construction also gives coefficient $31.928705\ldots$ (safe 32) for root degree at most $s^{3/5}$; a separate low-degree schedule gives $26.902328\ldots$ (safe 27) for root degree at most $s^{1/12}$. These strengthened-premise comparisons are proved separately below; they must not be conflated with the generic row.

A smaller coefficient gives a strictly smaller displayed exponential certificate, but comparisons of upper bounds do not determine the ratio of actual SAT runtimes. The kernel speedups in Section 8 are measured separately and have a narrower meaning.

1.1 What is and is not inherited

The source architecture deforms equality constraints, reconstructs by Hermite interpolation, extracts coefficients of separated crossing modes, and contracts smaller child networks. We retain its correctness identities, creation-time lifts, and permitted implicit-entry interface [3, 4]. We retain the new incoming asymmetric packing lemma and the full-block Cauchy decomposition after checking their compatibility [7, 8]. We do not rebuild the full upstream Lean development or claim that its scope covers these sharper constants.

The upstream scope page explicitly distinguishes its formalized general bound

$$(L_{\mathrm{enc}} + 1)^{O_k(1)} 2^{O_k(n(\log\log n)^2/\sqrt{\log n})}$$

from the October 25 manuscript's sharper

$$(n+m)^{O_k(1)} 2^{O_k(n\log\log n/\sqrt{\log n})}.$$

For fixed m/n the manuscript states $\mathrm{poly}(n,m)2^{O(n/\sqrt{\log n})}$ [5]. The deliberately admitted comparator challenge statement is not the proof endpoint; the public endpoint is `lean/OAI/Algorithms/SubexponentialSAT/SqrtLogSpace/Final.lean`. No claim here depends on mistaking a challenge file for a proof.

2 Pinned repository review

The initial review used snapshot `ab2fbc9e03d5`; a final delta review includes the extra matching packet and the subsequent placement. The final immutable snapshot is

`474ddba696a5d385c091c23fb2995f69edd13e00`.

Its timestamp is October 26, 2026, 7:11:13 PM Pacific (October 27, 02:11:13 UTC). The relevant commit sequence is:

Commit	Relevant change
`a505bd768bce`	First three ETH packets and two manuscript PDFs arrive.
`64b3202ee84f`	Precision, scalar-parity, and density refinements arrive.
`8afdbbea48d6`	Asymptotic-analysis and barrier packet arrives.
`2b925d8800cd`	Seven packets are placed under SubexponentialSAT.
`3e096089911f`	OpenAI manuscript sources are vendored unchanged.
`45233238aebf`	Project README summarizes the first seven packets.
`3e57705cf14e`	Seven additional ETH archives enter docs/incoming.
`d098ba436b33`	An eighth second-round packet, matching refinement, arrives.
`6d3adc450615`	All eight second-round packets are placed under SubexponentialSAT.

The reviewed OpenAI preprint is the version pinned in ProveIt at `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. Some packets use `42907c9e999c`; the relevant manuscript directory is recorded as byte-identical. The archive records the actual paths and hashes used, rather than relying on the moving branch name.

2.1 The seven placed packets

Packet	Established result claimed by the packet	Limit relevant here
`cauchy_mode_improvement`	Factorial Cauchy weights $1+3q-2\,\mathrm{popcount}(q)$, reducing leading mode entropy by $1/\sqrt{3}$.	Same logarithmic order.
`eth_contraction_improvements`	Sharp tapered condition counts; leading precision $6MD/u$; explicit coefficient 648.4411; streamed extraction.	Constant and component improvements.
`3sat_precision_refinement`	Exact loss budget and taper minimax; precision $24MD/u \to 6MD/u$.	Denominator-$\log n$ rate requires missing structured summation.
`sat_density_improvement`	Uniform density envelope $(n/\sqrt{\log n})\min\{\log(2+m/n), \log\log n\}$.	Worst-case logarithmic order unchanged.
`eth_density_refinement`	Overlapping density refinement; local odd-unit Cauchy Smith valuations.	No full all-level schedule improvement.

Packet	Established result claimed by the packet	Limit relevant here
`3sat_refinement`	Scalar parity lift, incidence cap $k(p+a-1)$.	Stronger recursive rate requires an unsupplied factorization.
`3sat_asymptotic_analysis`	Scoped taper/Newton barriers; identifies linear-wire and degree-preserving interfaces.	Neither missing interface is constructed.

These are reviewed as mathematical claims with stated dependencies, not promoted to formally verified results by their repository placement. The separate Claude manuscript has the same relevant unrestricted order $O(k^2 n \log\log n/\sqrt{\log n})$, also giving a clause-based branch $O(km/\sqrt{\log m})$. Its reported Lean development is not publicly linked in the filed manuscript. A targeted read of its Theorem 1 and Corollary 11 confirms that the constant is unspecified, and the statement uses a word-RAM model. Appendix E also says its formalization constant is not explicit. Therefore we do not claim numerical domination of that different algorithm or its uniform dependence on k. Our certificate comparison is among the explicit compatible OpenAI-contraction bounds; the kernel benchmark is narrower still.

2.2 The eight second-round packets

Packet	Main relevant contribution	Displayed generic coefficient
`3sat_round2_refinement`	Combined precision, factorial modes, packing and different arities at root and later calls.	84.0663; polynomial space.
`eth_packet_refinement`	Improved packet padding and schedule analysis.	95.6639; polynomial space.
`eth_balanced_packing`	Deterministic near-uniform pair loads, additive precision buffer.	94.7482; polynomial space.
`eth_high_moment`	High-moment/orbit packing and reduced labeling enumeration.	94.7482; polynomial space.
`eth_recursive_refinement`	Smaller fresh-side degrees and residue-aware 21-pivot modes with slope 64/21.	133.2074; polynomial space.
`eth_asymmetric_packet_balancing`	Asymmetric retained/fresh packing, sharper schedule, optional annulus bases.	76.9090; polynomial space.
`eth_matching_refinement`	Matching-based rational potential, near-uniform pair loads, and thin additive precision gaps.	94.7482; polynomial space.
`eth_joint_cauchy`	Full-orientation slope-four modes; explicit cached three-block summation using Strassen.	319.6858 polynomial space; 225.5971 time/196.1994 space.

The higher-moment packet additionally gives $42.3726842\ldots$ for root degree $s^{1/2}\operatorname{polylog} s$ and $31.2364960\ldots$ for bounded-degree roots. The recursive packet gives $60.7069575\ldots$ after a regrouping to degree $s^{0.51}$. The asymmetric packet's $31.0641321\ldots$ uses root degree $s^{1/12}$. The new degree-sensitive corollaries improve these numerical certificates under matched wire, precision, truncation, and oracle assumptions. They are included to avoid comparing different degree premises silently.

A packet's numerical coefficient is not its only reusable contribution. The joint Cauchy packet has a weaker standalone schedule but the stronger all-level mode basis. Combining that basis with asymmetric packing improves the frontier. Conversely, listing every conditional target as though it were an implemented upper bound would make the comparison invalid. The complete source

manifest accompanies the report. The final delta review matched 180 available text and code files of the original seven incoming packets to their new Git blob identities. The additional matching packet independently reaches coefficient $94.7482022595\ldots$ [10] using a matching potential and a thin-gap schedule. Its code covers those components and abstract mode counts; it supplies no full Cauchy recurrence or end-to-end SAT implementation. It therefore does not change the strongest frontier or the scope of the new kernel comparison.

2.3 The strongest order envelope and the conditional frontier

For fixed k the strongest established order among the reviewed reductions is

$$T(n,m) \le \mathrm{poly}_k(n,m)2^{O_k\left(\frac{n}{\sqrt{\log n}}\min\{\log(2+m/n),\log\log n\}\right)}. \tag{1}$$

At $m=\Theta(n\log\log n)$ this has exponent $O_k(n\log\log\log n/\sqrt{\log n})$, while unrestricted density retains $O_k(n\log\log n/\sqrt{\log n})$. Our new contraction certificates can be inserted into these reductions; they do not change the powers of the logarithms.

The strongest unproved summation proposal would replace the denominator $\sqrt{\log n}$ by $\log n$, giving $O(n\log\log n/\log n)$ in the exponent. Other missing-interface routes give $O(n(\log\log n)^{3/2}/\log n)$, $O(n\sqrt{\log\log n/\log n})$, or a linear-wire reduction. This report does not solve those interfaces and does not claim to beat their conditional asymptotic rates. It improves proved resource certificates and a concrete common kernel.

3 Why the algebraic ingredients compose

The compatibility check has four parts. Let M be the wire count at a call, $L=2^a$ the packet size, r the grouping alphabet, $Q=r^a$ its label count, and P the padded wire mass.

First, asymmetric packing provides $P=(1+o(1))M$ when $QR=o(M)$, $Q^2C(\log s)^2=o(M)$ and $LQ^2=o(M)$, where R,C bound the two sides' degrees. Below the root, retained tensors have degree at most the original invariant, whereas newly created packet tensors have degree at most $2L$ from the preceding level. This distinction permits a larger cumulative packet exponent.

Second, a full crossing orientation has one source group and one target group. Its feature functions may read all slots of a half-packet, but never cross the group boundary needed by the child-factorization identity. The real wires are unchanged and the fresh degree bound remains $2L$.

Third, each unequal-symbol packet has two orientations. At valuation level i, each mode index q has 2^i modes per orientation. Excluding equal symbols gives the family count

$$c_i \le (2(1-1/r)+o(1))M/D_i, \qquad D_i = L/2^i.$$

For a single directed group pair it is $(2/r^2+o(1))M/D_i$. These are the multiplicities required by both new resource certificates.

Finally, the sharp recovery budget remains $p_i \le (6+o(1))MD_i/u$ with the additive precision gap. The exact full-block weights are

$$w_q = a+1-i+4q-2\,\mathrm{popcount}(q).$$

Thus the mode replacement improves entropy without increasing child size, family multiplicity, or leading precision. The appendix inequalities below check all three resource constraints rather than multiplying unrelated published constants.

For perspective, using the same asymmetric $r = 16$ schedule and merely switching from slope three to four gives coefficient $76.90899495\sqrt{3/4} = 66.60514341\ldots$. Optimizing its continuous root/tail allocation at the strict packing boundary gives the comparison infimum $66.19272369\ldots$. For $r = 16$, the boundary root budget is $1/80$ and the remaining budget is $1/80$. Hölder's schedule minimization gives $K(16)\sqrt{80}(1 + 1/(3\sqrt{3})) = 66.19272369\ldots$. Strictly feasible schedules approach this value with slack. The multistage schedule below improves even that composition.

4 A polynomial-space certificate using several grouping arities

The previous round-two packet already permits different grouping alphabets at the root and below it. We therefore do not claim variable arity itself as a new idea. The additional refinement here is a concrete composition of that freedom with the asymmetric packing theorem and the full crossing-block Cauchy basis. In the same normalized Boolean tensor interface, it gives

$$T(s, \mathcal{L}) \le (s + \mathcal{L})^{O(1)} 2^{(C_{\mathrm{multi}} + o(1))s/\sqrt{\log_2 s}}, \qquad C_{\mathrm{multi}} = 63.85740947110328\ldots,$$

using polynomial space. Consequently the coefficient 64 is valid for all sufficiently large s. This is a sharper exponential upper-bound certificate. Its logarithmic order is unchanged, and its coefficient is per normalized tensor wire budget, not per SAT variable. The separate cached construction uses more space and has its own smaller time bound.

4.1 The inherited interface

For definiteness, assume that the root has at most s Boolean wires, maximum degree at most $\lceil s^{9/10} \rceil$, and initial coefficient precision and truncation order $O(s)$; use the same permitted entry-oracle cost and creation-time representative convention as the reviewed contraction theorems. Let $\lambda = \lceil \log_2 s \rceil$ and stop recursion at $H = \lceil s/\lambda \rceil$ wires.

At an active call with M wires, packet length $L = 2^a$, alphabet size r and $Q = r^a$ labels, the asymmetric packing lemma supplies padded mass $P = (1 + o(1))M$ provided, uniformly,

$$QR = o(M), \qquad Q^2 C\lambda^2 = o(M), \qquad LQ^2 = o(M).$$

Here R, C are the two side-degree bounds. At the root both are at most $\lceil s^{9/10} \rceil$. Below the root, orient retained tensors on the row side and fresh packet tensors on the column side. The bounds then are

$$R \le \lceil s^{9/10} \rceil, \qquad C \le 2^{a_{d-1}+1}.$$

The child-size ratio is $(2/r)(1 + o(1))$. For each Cauchy mode index there are at most

$$\big(2(1 - 1/r) + o(1)\big)M/D_i$$

mode families. The full-block weights are $a + 1 - i + 4j - 2s_2(j)$, and the sharp Hermite recovery budget remains $p_i \le (6 + o(1))MD_i/u_d$. The ordinary weighted-partition estimate therefore gives the following base-two logarithmic local cost:

$$\log_2 N_d \le \big(K(r_d) + o(1)\big)\frac{M_d}{\sqrt{u_d}}, \qquad K(r) = \frac{\pi}{\ln 2}\sqrt{2(1 - 1/r)}. \tag{2}$$

Indeed, a slope-four family contributes $(\pi^2 + o(1))/(24\theta)$ to its logarithmic generating function. Optimizing $\theta p_i + c_i\pi^2/(24\theta)$ gives $\pi\sqrt{2(1 - 1/r)}M/\sqrt{u_d}$ in natural logarithms. Neither the family count nor the source/target locality changes when r changes between calls. The entropy remainder is uniform because $D_i \ge \lambda^6$, $a = O(\lambda)$, and consequently $(M/D_i)\log^2(D_i/\sqrt{u_d}) = o(M/\sqrt{u_d})$.

4.2 An entirely explicit schedule

Put $b = \lceil 6 \log_2 \lambda \rceil$ and define

$$a_{-1} = 0, \qquad u_d = \lfloor f_d \lambda \rfloor, \qquad a_d = a_{d-1} + u_d + b.$$

Use the first four pairs

$$(r_0, r_1, r_2, r_3) = (13, 14, 8, 6), \qquad (f_0, f_1, f_2, f_3) = \left(\frac{27}{2000}, \frac{159}{12500}, \frac{71}{10000}, \frac{533}{100000} \right).$$

For every $j \geq 0$ use

$$r_{4+j} = 4, \qquad f_{4+j} = \frac{27}{6400} \left(\frac{5}{8} \right)^j.$$

The four cumulative leading exponents and their limiting total are

$$(A_0, A_1, A_2, A_3) = \frac{1}{100000}(1350, 2622, 3332, 3865), \qquad A_\infty = \sum_{d \geq 0} f_d = \frac{499}{10000}.$$

The asymmetry is useful because the root requires $2A_0 \log_2 r_0 + 9/10 < 1$, whereas a later row-side constraint is only $A_d \log_2 r_d + 9/10 < 1$. Smaller later alphabets allow a larger cumulative packet exponent without violating that constraint.

All required inequalities can be checked using rational upper bounds

$$\log_2 13 < \frac{3701}{1000}, \quad \log_2 14 < \frac{381}{100}, \quad \log_2 6 < \frac{1293}{500},$$

certified respectively by the integer comparisons $13^{1000} < 2^{3701}$, $14^{100} < 2^{381}$ and $6^{500} < 2^{1293}$. The resulting upper powers of s are as follows. Logarithmic factors, ceilings and precision buffers contribute only $s^{o(1)}$.

call	dominant high-degree term	$Q_d^2 C_d$	$L_d Q_d^2$
0	0.999927	0.999927	0.113427
1	0.9998982	0.2132964	0.2260164
2	0.99996	0.22614	0.23324
3	0.9999489	0.2332178	0.2385478
$d \geq 4$	0.9998	0.2495	0.2495

In the first row the dominant term is $Q_0^2 \Delta$; below it the high-degree term is $Q_d \Delta$. Every exponent has a fixed strict margin. The smallest displayed high-degree margin is $1/25000$. New packet degrees are $s^{499/10000 + o(1)}$, hence are much smaller than the maintained $s^{9/10}$ bound.

For a noncircular depth argument, provisionally check all depths up to $4 + \lceil \log_{20/11} \lambda \rceil + O(1)$. Their total additive buffer is $O((\log \lambda)^2)$, and their smallest u_d is

$$\lambda^{\eta + o(1)}, \qquad \eta = 1 - \frac{\ln(8/5)}{\ln(20/11)} > 0.$$

The positivity follows already from $5/8 > 11/20$. Thus flooring is a uniform relative $o(1)$ perturbation. The strict power margins make all packing conditions uniform throughout this provisional depth range. For sufficiently large s every child has at most $(11/20)M$ wires, because its limiting ratio is at most $1/2$. An active path therefore cannot exceed the provisional depth. This closes the bootstrap.

The precision argument with additive gaps is unchanged: inherited precision is $p = O(s\lambda^2 2^{a_{d-1}})$, while $D_i \geq 2^{a_{d-1} + b + 1}$ for $i < u_d$. Since $M > s/\lambda$ and $u_d \leq \lambda$, inherited precision contributes at most $O(\lambda^4 2^{-b}) = o(1)$ times $M D_i / u_d$. The extra jet-loss term is also negligible, so the coefficient six used in (2) is uniform.

4.3 The coefficient and the remaining costs

Ignoring only uniform relative $o(1)$ perturbations, the size multipliers at the first five depths are

$$m_0 = 1, \quad m_1 = \frac{2}{13}, \quad m_2 = \frac{2}{91}, \quad m_3 = \frac{1}{182}, \quad m_4 = \frac{1}{546}.$$

Thereafter $m_{4+j} = m_4 2^{-j}$. Therefore

$$C_{\text{multi}} = \frac{K(13)}{\sqrt{27/2000}} + \frac{2K(14)}{13\sqrt{159/12500}} + \frac{2K(8)}{91\sqrt{71/10000}} + \frac{K(6)}{182\sqrt{533/100000}} + \frac{K(4)}{546\sqrt{27/6400}} \frac{1}{1-\sqrt{2/5}}.$$

The tail converges because $\sqrt{2/5} < 1$. Uniform padding errors perturb a strictly convergent geometric series by $o(1)$. The provided exact checker proves $C_{\text{multi}} < 64$ using only rational arithmetic, upper square-root enclosures, $\pi < 22/7$ and $\ln 2 > 693/1000$; these deliberately coarse enclosures already give an upper bound below 63.897.

Polynomial interpolation and grouping factors at $O(\log \lambda)$ depths add $O(\lambda \log \lambda)$ to the logarithm of the work. Direct leaves cost $2^{O(s/\lambda)}$. The permitted original entry cost is $\exp(O(s^{9/10}))$, and packet entry extraction costs at most $\exp(O(L \log s)) = \exp(s^{499/10000+o(1)})$. All are $2^{o(s/\sqrt{\lambda})}$. Labels, interpolation degrees, precisions, and stored immutable descriptions remain polynomial in the normalized input size; depth-first enumeration therefore preserves polynomial space. This proves the stated certificate relative to the unchanged source identities and oracle interface.

The packet does not assert that this multistage schedule is globally optimal. Its purpose is a fully checkable strict improvement over the fixed-arity polynomial-space composition. It neither establishes the conditional $n/\log n$ proposals nor removes the $\log \log n$ factor in the general-density reduction.

5 Preserving the stronger degree-sensitive frontier

A comparison restricted to the generic $s^{9/10}$ degree interface would miss useful stronger-premise bounds in the reviewed packets. We record two polynomial-space corollaries on the same normalized wire scale. Both retain the same permitted original entry-oracle cost as the generic theorem. Neither changes the unrestricted SAT logarithmic order.

5.1 Root degree at most $s^{3/5}$

Suppose the root degree is at most $\lceil s^{3/5} \rceil$. Keep the arities from Section 4 and multiply every leading retained-level coefficient f_d by

$$\tau = 4.$$

Continue using $u_d = \lfloor \tau f_d \lambda \rfloor$ and $a_d = a_{d-1} + u_d + \lceil 6 \log_2 \lambda \rceil$. Maintain the degree bound $\lceil s^{3/5} \rceil$ on the retained side. The cumulative packet exponent is now

$$\tau A_\infty = \frac{499}{2500} = 0.1996 < \frac{3}{5},$$

so every fresh packet tensor satisfies the maintained bound eventually.

The high-degree packing powers become $3/5 + \tau h_d$, where h_d denotes the label-exponent part of the corresponding bound in Section 4; their maximum is

$$\frac{3}{5} + 4\frac{2499}{25000} = \frac{6249}{6250} = 0.99984 < 1.$$

The root uses the doubled label exponent as before. All pure label, fresh-column and rounding powers are multiplied by τ; their maximum is at most

$$4\frac{499}{2000} = \frac{499}{500} = 0.998 < 1.$$

Thus all packing conditions retain fixed strict margins. The same coarse child shrinkage, retained-range ratio and depth argument apply. Multiplying every u_d by a fixed positive factor also leaves the additive precision-gap proof unchanged.

The local costs are divided by $\sqrt{\tau}$. Consequently this interface has polynomial-space time bound

$$(s + \mathcal{L})^{O(1)} 2^{(C_{3/5}+o(1))s/\sqrt{\log_2 s}}, \qquad C_{3/5} = \frac{C_{\mathrm{multi}}}{2} = 31.92870473555164\ldots.$$

The integer coefficient 32 is valid eventually: halve the earlier rational upper bound on C_{multi} to obtain a bound below 31.949. Packet-entry work is now at most $\exp(s^{0.1996+o(1)})$, still lower order; all other resource arguments remain as in the generic theorem. An optional schedule with larger power margins uses $\tau = 39/10$ instead; it gives coefficient $32.33545622568384\ldots$ and safe integer coefficient 33.

This premise includes root degree $s^{1/2}\,\mathrm{polylog}(s)$ and root degree $s^{0.51}$ for sufficiently large s. Subject to their common normalized precision, truncation and oracle hypotheses, it therefore improves the respective $42.3726842\ldots$ and $60.7069575\ldots$ degree-sensitive polynomial-space certificates from the incoming high-moment and recursive packets. This comparison is between upper bounds on the same normalized network size.

5.2 Root degree at most $s^{1/12}$

For low-degree input it is better to compose the incoming asymmetric packet's already proved low-degree schedule with the incoming full-block Cauchy basis. Maintain $\Delta = \lceil s^{1/12} \rceil$, and use

$$r = 48, \qquad u_d = \left\lfloor \frac{\lambda}{14\,8^d} \right\rfloor, \qquad a_d = \sum_{j=0}^{d} \left(u_j + \lceil 6 \log_2 \lambda \rceil\right).$$

The leading cumulative packet exponent is at most $4/49$. The exact integer inequality $3^5 < 2^8$ yields $\log_2 48 < 28/5$. Therefore the symmetric degree-packing exponent is bounded by

$$2\frac{4}{49}\log_2 48 + \frac{1}{12} < \frac{419}{420} < 1,$$

and the rounding exponent is bounded by

$$(1 + 2\log_2 48)\,\frac{4}{49} < \frac{244}{245} < 1.$$

The new packet degree has leading exponent $4/49 < 1/12$, so the same small degree bound is preserved. The coarse child ratio $1/12$ and retained-range ratio $1/8$ give the usual noncircular depth bootstrap, with a uniformly positive power of λ in every active u_d.

The sharp child ratio is 1/24. The slope-four local coefficient is

$$K(48) = \frac{\pi}{\ln 2}\sqrt{\frac{47}{24}}.$$

Summing the convergent geometric series gives

$$C_{1/12} = \frac{K(48)\sqrt{14}}{1-\sqrt{8}/24} = \frac{\pi\sqrt{329/12}}{\ln 2\,(1-\sqrt{2}/12)} = 26.90232751306742\ldots.$$

Thus the polynomial-space bound has eventual integer coefficient 27. The exact checker verifies this directly using rational enclosures. This improves the prior polynomial-space coefficients $31.0641321\ldots$ and $31.2364960\ldots$ on their overlapping low-degree premises. In particular, bounded-degree clause networks eventually satisfy the $s^{1/12}$ premise. Conversion to a SAT-variable coefficient still requires accounting for the chosen encoding's wire budget.

These two corollaries are compositions with complete schedules and resource accounting. They do not imply numerical dominance between networks measured in different wire parameters, and they do not compare polynomial space with an unrestricted-memory claim as though the two resource guarantees were identical.

6 An optional faster contraction with subexponential space

The two strongest compatible ingredients in the new incoming packets address separate interfaces. Asymmetric packet balancing changes the number and sizes of child networks. The full-orientation Cauchy decomposition changes their coefficients while preserving the same wires and groups. The latter packet also supplies an explicitly costed caching algorithm. Composing these three ingredients, and choosing a different alphabet and depth schedule, gives the following additional time–space tradeoff. The algorithmic primitives are credited to the reviewed packets; the composition and its resource certificate are the contribution of this section.

Theorem 6.1 (Cached contraction with eight groups)**.** *Use the normalized input interface of the reviewed contraction reports: a bipartite Boolean tensor network has at most s wires, maximum degree at most $\lceil s^{9/10}\rceil$, and description length $\mathcal{L}$. Its entries lie in $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^{\ell})$, with $P_0, \ell \le Cs$ for a fixed C. Original tensor entries are evaluable in $(s+\mathcal{L})^{O(1)}2^{O(s^{9/10})}$ time and polynomial space. Retain the source's exact deformation, Hermite recovery, child identity, and immutable creation-precision convention. Then the composed algorithm has bit time*

$$(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{time}}+o(1))s/\sqrt{\log_2 s}}$$

and bit space

$$(s+\mathcal{L})^{O(1)}2^{(C_{\mathrm{space}}+o(1))s/\sqrt{\log_2 s}},$$

where, putting $\omega = \log_2 7$,

$$C_{\mathrm{time}} = \frac{\pi\sqrt{122}}{8\ln 2}\sqrt{30+12(\omega-2)^2} = 38.48449590573546\ldots,$$
$$C_{\mathrm{space}} = \frac{\pi\sqrt{3660}}{8\ln 2} = 34.27480300502269\ldots.$$

In particular, 39 and 35 are valid eventual coefficients. These are coefficients per normalized tensor-wire budget, not per SAT variable, and are upper-bound comparisons rather than measured runtime ratios.

Schedule and packing. Let $\lambda = \lceil \log_2 s \rceil$, let $H = \lceil s/\lambda \rceil$, and set

$$r = 8, \qquad b = \lceil 6 \log_2 \lambda \rceil, \qquad u_d = \left\lfloor \frac{\lambda}{61\, 2^d} \right\rfloor, \qquad a_d = \sum_{j=0}^{d} (u_j + b), \quad a_{-1} = 0.$$

A call with at most H edges is a direct-summation leaf. At an active call of depth d, put $L_d = 2^{a_d}$ and $Q_d = 8^{a_d}$. At the root both side-degree bounds are $\Delta = \lceil s^{9/10} \rceil$. Below the root, orient the bipartition with all retained tensors on the row side and all new packet tensors on the column side. Then

$$R_d \le \Delta, \qquad C_d \le 2L_{d-1}.$$

The near-unit padding lemma in the asymmetric report applies when $Q_d R_d / M = o(1)$, $Q_d^2 C_d \lambda^2 / M = o(1)$, and $L_d Q_d^2 / M = o(1)$. To check these conditions without assuming the conclusion about recursion depth, first consider hypothetical depths $d \le \lceil \log_3 \lambda \rceil + 2$ and active $M > s/\lambda$. Uniformly there,

$$a_0 \le \lambda/61 + O(\log \lambda), \qquad a_d \le 2\lambda/61 + O((\log \lambda)^2), \qquad u_d = \Omega(\lambda^{1-\log_3 2}).$$

At the root the largest degree constraint has exponent $6/61 + 9/10 = 609/610 < 1$. Below the root, the retained-row constraint has that same exponent, while both the fresh-column and rounding constraints have exponent at most $14/61 < 1$. Thus every relative excess in the padding lemma tends uniformly to zero. The padded mass is $P = (1 + o(1))M$, and every child has at most

$$2P/r = (1 + o(1))M/4$$

wires. For all sufficiently large sizes this is at most $M/3$, which proves the hypothetical depth bound and closes the bootstrap. Also every newly created tensor has degree at most $2L_d = s^{2/61 + o(1)} < s^{9/10}$, preserving the degree invariant. The algorithm can check its finite parameter and padding inequalities exactly and use direct summation at exceptional bounded sizes.

Precision. At valuation level $i < u_d$, write $D = L_d/2^i$ and choose the sharp taper

$$h = \left\lceil \frac{M(2L_d - 1) + 1}{L_d(2u_d - 2 + 2^{1-u_d})} \right\rceil.$$

Use the previously justified working precision $p_t = p + 6hD + hu_d(a_d + 1)$. The additive gap gives $D \ge 2^{a_{d-1} + b + 1}$ and $p = O(s\lambda^2 2^{a_{d-1}})$. Since $M > s/\lambda$, it follows that

$$\frac{p}{MD/u_d} = O(\lambda^4 2^{-b}) = o(1), \qquad \frac{hu_d(a_d + 1)}{MD/u_d} = o(1).$$

Uniform taper growth also gives $h = (1 + o(1))M/u_d$. Consequently $p_t = (6 + o(1))MD/u_d$. All these integers, and all interpolation data, have polynomial bit length. In particular $M(2L_d - 1) < s^4$ eventually, so the unchanged residue-field choice supplies enough interpolation points.

Modes for each directed pair. The full-orientation Cauchy decomposition provides

$$w_q = a_d + 1 - i + 4q - 2\,\mathrm{popcount}(q), \qquad 0 \le q < D/2.$$

It has 2^i modes of each index q per crossing orientation. Its source form reads a whole half-packet, but that half-packet belongs to one source group; every target coefficient is used by one target group.

The child wires and fresh-tensor degrees are therefore unchanged. Uniform padding gives, for each directed pair $g \neq h$, at most

$$c_{gh} = \frac{2P}{r^2 D} = \left(\frac{2}{r^2} + o(1)\right)\frac{M}{D}$$

modes of each q. The factor $r(r-1)$, rather than r^2, counts the crossing pairs; there are no modes for equal symbols.

Enumerate exact per-pair weight arrays (b_{gh}) with total below p_t. There are $(p_t+1)^{56}$ possible arrays, a polynomial number because $r = 8$ is fixed. Each feasible mode tuple belongs to exactly one array. Exact-weight lists can be enumerated with polynomial delay and indexed using the standard count recurrence $A(j,t) = A(j-1,t) + A(j,t-w_j)$, with $A(0,0) = 1$ and the other empty-list counts zero. There are polynomially many modes and budgets; count bit lengths are polynomial as well. No completion-counting oracle is assumed.

Discard arrays with an empty pair list; they contribute zero. For every remaining exact array let h_e be the base-two logarithm of the pair-list size for directed pair e. Define

$$K(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

For any fixed nonnegative weights v_e, the same generating-product argument as in the joint Cauchy report yields

$$\sum_e v_e h_e \leq \left(\sqrt{\frac{\sum_e v_e^2}{r^2}} + o(1)\right) K(M,u). \tag{3}$$

Indeed, for $\widehat{w}_q = 1 + 4q - 2\,\mathrm{popcount}(q)$,

$$\sum_{q\geq 0} -\ln(1 - e^{-\theta\widehat{w}_q}) = \frac{\pi^2}{24\theta} + O(\log^2(2/\theta)).$$

Apply the product bound to pair e with parameter θ/v_e, sum after multiplying by v_e, and optimize a common θ. Using $\sum_e b_e < p_t$ and the displayed color count gives (3). This argument also covers tiny or zero per-pair budgets, without a separate saddle approximation for each. The remainder is uniform: $D \geq \lambda^6$ and $(M/D)\log^2(D/\sqrt{u}) = o(M/\sqrt{u})$.

Explicit tables and their costs. Cache the value of each group factor Z_g on all its incident pair-index tuples, paying for every value by recursively contracting the actual child. A group has 14 incident directed pairs, so this needs at most $2^{(a_0+o(1))K(M,u)}$ child calls per group, where $a_0 = \sqrt{14}/8$. Partition the eight groups into blocks of sizes $3, 3, 2$. A block of size k has $2rk - k^2 - k$ incident directed pairs and $2k(r-k)$ boundary pairs. The maximum counts are 36 and 30. Literal summation of its internal indices, using the cached group values, therefore builds all three boundary tables in time $2^{(a_B+o(1))K}$ and space $2^{(j_B+o(1))K}$, where

$$a_B = 3/4, \qquad j_B = \sqrt{30}/8.$$

The table-building enumeration does not make new recursive oracle calls.

The three block-pair index classes contain $18, 12, 12$ directed pairs. Writing their log-dimensions as H_1, H_2, H_3, division-free rectangular Strassen multiplication evaluates the triangle of tables in time with exponent

$$H_1 + H_2 + H_3 - (3-\omega)\min(H_1, H_2, H_3), \qquad \omega = \log_2 7.$$

This is a maximum of three weighted sums. Inequality (3) bounds its coefficient by

$$b_S = \frac{\sqrt{30 + 12(\omega - 2)^2}}{8} = 0.768743532920437\ldots.$$

The largest bound discounts one of the size-12 classes. Since $a_B < b_S$, this matrix step dominates table construction. Strassen uses only ring addition, subtraction, and multiplication and is valid modulo powers of two, including with nilpotents. Square-block processing retains $O(d^2)$ elements for a block of side d; these fit inside the already bounded boundary-table storage.

These are bit-operation bounds, not unit-cost random-access assumptions. Caches can be ordered by their projected lexicographic axis order; suffix rewinds are charged to the full assignments just traversed. Reblocking and transposing a boundary table by sorting records costs its size times polynomial factors in the record bit length and logarithm of its size. Those operations stay below the displayed time exponent. All pair axes, ring elements, and exact-weight indices have polynomial-size records.

The entire recursion. Up to lower-order polynomial costs, the work recurrence is

$$T_d \le 2^{(a_0+o(1))K_d} r T_{d+1} + 2^{(a_B+o(1))K_d} + 2^{(b_S+o(1))K_d}.$$

The schedule and child bound imply

$$K_d \le (1 + o(1))\frac{\pi\sqrt{122}}{\ln 2}\frac{s}{\sqrt{\lambda}}\gamma_s^d, \qquad \gamma_s \longrightarrow \gamma = \frac{\sqrt{2}}{4}.$$

The coefficient incurred by preceding child-cache enumerations and the matrix work at depth d is at most

$$a_0\frac{1 - \gamma_s^d}{1 - \gamma_s} + b_S\gamma_s^d + o(1).$$

Now $a_0/(1 - \gamma) = 0.7235047203294623\ldots < 3/4 < b_S$. Thus the root matrix coefficient b_S bounds every depth, including an eventual leaf. Geometric sums are stable under the uniform perturbation $\gamma_s \to \gamma$, so no unproved exchange of an infinite-depth limit is needed. Polynomially many interpolation and exact-weight choices at $O(\log \lambda)$ depths contribute $O(\lambda \log \lambda)$ additional log-work. Leaves cost $2^{O(s/\lambda)}$; original entry routines and new packet entries cost $2^{O(s^{9/10})}$ and $\exp(s^{2/61+o(1)})$, respectively. All are lower order. A fixed polynomial in $s + \mathcal{L}$ is charged per visited call, rather than raised recursively to a description-dependent exponent.

Depth-first processing retains only ancestor tables, the current tables, and polynomially sized descriptions and interpolation data. Since K_d decreases geometrically, their total storage has leading coefficient $j_B\pi\sqrt{122}/\ln 2$. This proves the two stated constants. Exact elementary bounds $\pi < 22/7$, $\ln 2 > 6931/10000$, and $\log_2 7 < 281/100$ certify the rounded eventual coefficients 39 and 35.

Corollary 6.2 (A lower-degree specialization)**.** *If the root degree is at most* $\lceil s^{1/12} \rceil$*, keep* $r = 8$ *but replace* 61 *by* 15 *in the depth schedule. Maintain degree bound* $\Delta' = \lceil s^{1/7} \rceil$*. New packet degrees are* $s^{2/15+o(1)} < s^{1/7}$*, and even the symmetric packing bound has exponent* $6(2/15) + 1/7 = 33/35 < 1$*. Hence the same proof applies, with time and space coefficients*

$$\frac{\pi\sqrt{30}}{8\ln 2}\sqrt{30 + 12(\omega - 2)^2} \quad \textit{and} \quad \frac{30\pi}{8\ln 2},$$

respectively. Original entry routines may retain their allowed $2^{O(s^{9/10})}$ *cost. This corollary has a stronger graph-degree premise and must be compared to the earlier low-degree certificates separately.*

Interpretation. This composition preserves the source's exact-counting semantics but uses subexponential space. It does not replace the polynomial-space result. It does not close the prior structured-summation or compact-factorization hypotheses, does not improve the established logarithmic order in n, and does not provide an end-to-end SAT implementation. Its scope is a stronger, fully accounted normalized contraction certificate, conditional only on the same unchanged source framework as the reviewed refinements.

7 Quadratic-time full Cauchy feature tables

This improvement has an independently executable scope. It computes the same two separated feature tables as the incoming joint Cauchy prototype, modulo 2^p, without reconstructing every prefix product from scratch. The algebra holds over any unramified two-adic coefficient ring. The supplied implementation specializes to

$$\mathcal{R}_p = (\mathbb{Z}/2^p\mathbb{Z})[U]/(U^2 + U + 1),$$

including non-prime-field interpolation values. It is a local kernel; it does not implement the complete tensor contractor or the SAT reduction.

7.1 The exact product formulas

Let $L = 2^a$, $R = 2^i$, $N = L/2$, and

$$s_j = (j \bmod R) + 2R\lfloor j/R \rfloor, \qquad T = t + 2^{a+1}\zeta, \qquad v_2(t) = i < a,$$

where $-L \le t < L$ and ζ is integral. Define

$$E_b = 2\lfloor b/R \rfloor - \mathrm{popcount}(\lfloor b/R \rfloor), \qquad \mathcal{C}(T) = \prod_{q=-L}^{L-1} (T - q), \qquad B = \frac{\mathcal{C}(T)}{\mathcal{C}'(0)T}.$$

For $j \ge b$ the integral normalized factors are

$$F_b(j) = \frac{T}{T + s_j - s_b}\, 2^{-E_b} \prod_{h<b} \frac{s_j - s_h}{T + s_j - s_h}, \tag{4}$$

$$G_b(j) = \frac{T}{T + s_b - s_j}\, 2^{-E_b} \prod_{h<b} \frac{s_h - s_j}{T + s_h - s_j}. \tag{5}$$

Both are zero for $j < b$. The target table is $A_b(j) = B2^{2E_b}F_b(j)$; the source table is $G_b(j)$. The incoming finite Cauchy identity is

$$\frac{\mathcal{C}(T)}{\mathcal{C}'(0)(T + s_j - s_k)} = \sum_{b=0}^{N-1} A_b(j)G_b(k). \tag{6}$$

The normalized factors are integral and have unit diagonal. The target factor has valuation at least $a + 1 - i + 2E_b$. We reuse that identity rather than claim its discovery.

7.2 A telescoping update between consecutive modes

Lemma 7.1 (Incremental normalized features)**.** *For $0 \le b < j < N$,*

$$F_{b+1}(j) = F_b(j)\frac{s_j - s_b}{T + s_j - s_{b+1}}2^{-(E_{b+1}-E_b)}, \tag{7}$$

$$G_{b+1}(j) = G_b(j)\frac{s_b - s_j}{T + s_{b+1} - s_j}2^{-(E_{b+1}-E_b)}. \tag{8}$$

The initial values are $F_0(j) = T/(T + s_j)$ and $G_0(j) = T/(T - s_j)$.

Proof. Divide the $b+1$ product in (4) by the b product. The old leading denominator cancels the denominator of the new product factor. The remaining denominator is the new leading denominator. The normalization changes by $2^{-(E_{b+1}-E_b)}$. The same cancellation with reversed differences proves (8). The restriction $b < j$ avoids a zero numerator; entries after $b = j$ are known to vanish by triangularity. □

7.3 Finite precision requires unit states

An individual update ratio need not be integral, even though every output $F_b(j), G_b(j)$ is integral. Applying the recurrence to residues alone would therefore be wrong. Store a nonzero state as

$$(v, u), \qquad v \in \mathbb{Z}, \quad u \in \mathcal{R}_p^{\times}, \qquad x = 2^v u \pmod{2^{p+v}}.$$

Here the assertion means the exact value has valuation v and its unit part is known modulo 2^p; only when displaying an integral feature is $2^v u$ reduced modulo 2^p. Multiplying or dividing a ratio first removes its exact powers of two, then multiplies the stored unit by unit factors. No even element is inverted modulo 2^p.

Lemma 7.2 (Safe precision transport)**.** *If an exact nonzero state has known valuation v and unit part known modulo 2^p, every update above determines the next exact valuation and its unit part modulo 2^p. Whenever the next valuation is nonnegative, its output residue modulo 2^p is correct, including after an intermediate zero output residue.*

Proof. Write the update ratio as $2^e d/c$ with c, d units. Its unit inverse exists in $\mathcal{R}_p$, and the new representation is $(v + e, udc^{-1})$. Knowledge of the unit modulo 2^p is preserved under these ring operations. If $v + e \ge 0$, the omitted uncertainty is divisible by 2^{p+v+e}, hence by 2^p. No information has been discarded merely because an earlier displayed residue was zero. □

A concrete regression is $a = 3$, $i = 0$, $t = 1$, $\zeta = 1$, $j = 2$. The valuations of $F_0(j), F_1(j), F_2(j)$ are $0, 1, 0$. At precision one the middle residue is zero but the last is a unit. The supplied tests explicitly require recovery after that zero.

7.4 Only one unit inversion

All denominators have the form $T + \delta$, where

$$\delta \in \mathcal{D} = \{2Rq + v : 1 - N/R \le q < N/R,\ 1 - R \le v < R\}.$$

The intervals are disjoint, so $|\mathcal{D}| = (2N/R - 1)(2R - 1) < 2L$. They are exactly the differences of the missing-bit slots. For each $\delta \in \mathcal{D}$, $v_2(T + \delta) \le i$, with the same valuation as $t + \delta$. Integer lift

coordinates are normalized before modular reduction. Blindly using every integer $|\delta| < L$ would wrongly include $\delta = -t$; the actual difference set excludes that singular case.

Batch inversion works over any commutative ring for unit inputs $d_1, \dots, d_h$: form prefix products, invert their total once, and scan backward to recover every d_j^{-1}. It costs $O(h)$ multiplications and one inversion. We include the normalized odd scalar denominator of $\mathcal{C}'(0)$ in that same batch.

The prefactor is computed after cancelling T algebraically:

$$B = \frac{\prod_{q \neq 0}(T - q)}{\prod_{q \neq 0}(-q)}.$$

Again retain valuation and unit separately. For nonzero ζ its valuation is exactly $a + 1 - i + v_2(\zeta)$. If $\zeta = 0$, the prefactor and every target feature are zero, but the source features are still computed correctly. This case is tested separately.

Theorem 7.3 (Table construction complexity)**.** *The two dense $N \times N$ feature tables can be constructed in $O(N^2 + L)$ ring operations and one unit inversion. The table stage of the supplied implementation uses exactly $(5N^2 + N)/2$ ring multiplications and $N(N-1)$ recurrence transitions. In comparison, direct construction of the product formulas uses $(N^3 - N)/3$ inner product steps across the two tables, hence $\Theta(N^3)$ ring work.*

Proof. At each fixed j generate the $j + 1$ nonzero entries in each table by Lemma 7.1. There are $N(N-1)$ transitions total. Each transition uses two ring multiplications. The two initial states per column use $2N$ products; forming each target feature uses another $N(N+1)/2$. Their sum is $(5N^2 + N)/2$. The preparation scans $O(L)$ factors and denominators. For the direct formulas, sum $2\sum_{b=0}^{N-1} b(N-b) = (N^3 - N)/3$. □

The dense output itself has $\Theta(N^2)$ entries, so quadratic work is optimal at the ring-operation output scale. This is not a claim about optimal compressed representations or computing only a few entries. Let $\mathsf{M}(q)$ denote a q-bit multiplication cost, $\mathsf{I}(p)$ one modular inverse, and B_{in} the maximum bit length of an exact input coordinate. A safe bit bound for this implementation is

$$O\big((N^2 + L)\mathsf{M}(p + a) + LB_{\text{in}} + \mathsf{I}(p)\big),$$

with space $O(N^2 p + L(p+a) + B_{\text{in}})$ bits. This accounts for exact normalization and index arithmetic even when $p = 1$. For $p \geq a$ and $B_{\text{in}} = O(p + a)$ it simplifies in the usual way. The coefficient-ring operations in the general contraction theorem have their own polynomial representation overhead.

8 Exact validation and measured performance

8.1 Correctness evidence

The final kernel checks record all of the following:

- 1,200 parameter cases comparing the incremental implementation with a direct-product modular implementation; 273,120 feature entries.
- 138,288 complete or sampled matrix reconstruction checks using (6), including 36 larger stress cases.

- 100 parameter cases and 6,260 feature entries compared with the historical exact-rational `joint_cauchy.py` implementation.
- Precision-one recovery after a transient zero, zero prefactors, negative interpolation nodes, non-prime-field values, high-valuation values, and precisions through 129 bits.

The direct modular comparator shares the ring and denominator cache, so it alone would not exclude a common arithmetic error. The historical implementation uses exact `Fraction` arithmetic in a quadratic field and supplies an independent arithmetic path. Matrix reconstruction checks the defining kernel independently of the individual feature comparison. The proof explains all parameter sizes; finite tests supplement that proof rather than replace it.

The three resource-certificate checkers use exact rational inequalities, integer comparisons for logarithms, and outward square-root bounds. Verification gates remain active with `python -O`; both modes pass. The polynomial-space checker records honest failures of its conservative finite-size diagnostic at $\log_2 \lambda = 20, 24, 28, 32$ and passes at $40, 48, 64$. Those very large thresholds illustrate the asymptotic nature of these schedules; they are not evidence of a usable SAT solver at ordinary sizes.

8.2 Matched modular timing

The benchmark uses CPython, $p = 64$, $\zeta = U$, complete materialization of both tables, and the same cached denominators for both methods. Construction costs are included, execution order alternates, and each row is the median of five samples. Outputs are checked for equality in every sample. Machine and Python details, all samples, parameters and operation counters are in `results/benchmark_results.json`.

N	Direct modular (ms)	Incremental (ms)	Speedup
8	0.403	0.253	1.59
16	2.017	0.743	2.71
32	13.867	2.331	5.95
64	99.609	8.540	11.66
128	791.950	30.266	26.17

At $N = 128$, the median drops from approximately 0.7920 seconds to 0.03027 seconds, a measured 26.17-fold reduction. The trend is consistent with the separately proved cubic-to-quadratic work reduction. Timings on one shared machine do not establish universal speedup ratios.

The historical exact-rational implementation was also measured, including conversion of its output to the same modular ring. At $N = 32$, its median is 1.0034 seconds against 0.002017 seconds for the new method, a 497.39-fold ratio. That larger ratio includes both the arithmetic representation change and prefix reuse; it should not be attributed solely to the recurrence. Both benchmark series are retained.

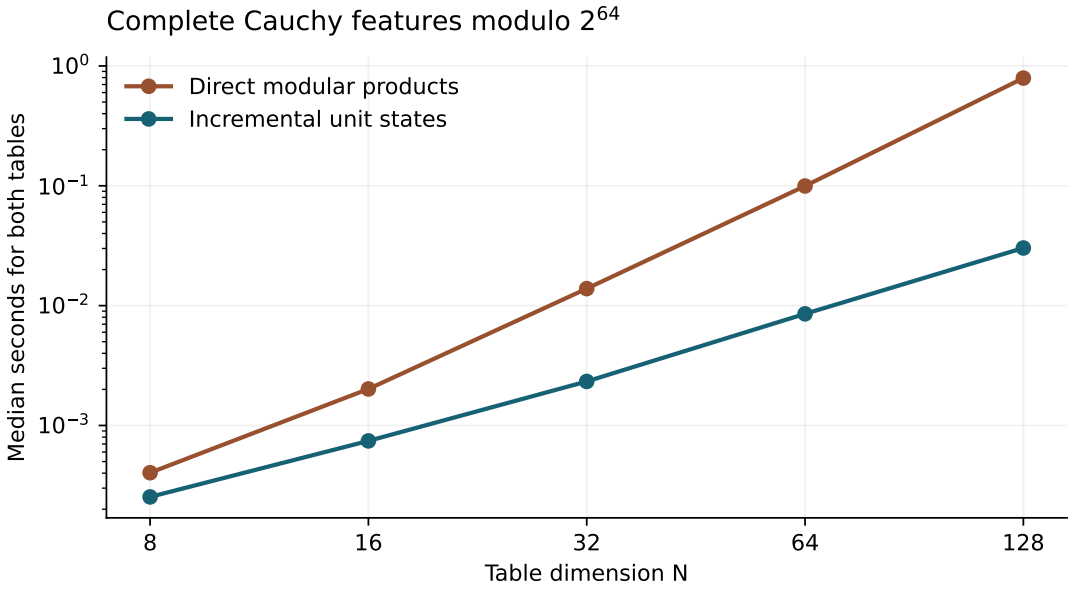


Figure 1: Measured time for complete two-table construction with identical 64-bit output. The direct modular baseline isolates prefix reuse; these are local kernel measurements, not SAT timings.

9 Integration, limitations, and further research

9.1 A reviewable integration path

The archive is a self-contained research packet. It supplies this PDF and its complete LaTeX sources; the modular kernel; a matched direct-product baseline; the unmodified historical Cauchy prototype and its repository license; independent checks; exact schedule certificates; all measured samples; and source pins and hashes. It makes no changes to the remote repository. It can be filed under the existing research directory or submitted through the user's usual incoming process.

For kernel integration, preserve the public meaning of the two feature tables and compare residues at the caller's precision. Keep immutable integer input lifts and valuation/unit states. The current module's ring is quadratic; a production contractor must connect the same recurrence to its actual unramified coefficient-ring interface. A large-field implementation must preserve normalization before reduction, rather than simply casting the quadratic prototype to another modulus.

For the polynomial-space bound, use the stated multistage arities and retained-level fractions with the additive gap. For the optional cached bound, use the separate eight-group schedule and account for its subexponential storage. These are distinct resource choices. A finite implementation should check every padding and precision inequality and use the exact direct algorithm whenever the required finite inequalities fail.

9.2 What this iteration leaves open

The report establishes an improved explicit contraction certificate and a faster exact local kernel. It does not establish a better unrestricted logarithmic order, a new end-to-end SAT implementation, a smaller numerical coefficient per SAT variable, or an independent formal proof of the source architecture. The optional cached construction spends additional space. All of these distinctions

matter to a fair comparison with the fifteen packets.

The missing-interface proposals cannot be converted to theorems merely by giving mode assignments shorter names. One must preserve group-local structure, bound resulting tensor degrees, implement the sum, and pay for all entry-oracle and bit operations. Similarly, exact Smith valuations are a local constraint on changes of basis for a full padded block, not a lower bound on SAT algorithms.

9.3 Specific further questions

1. **Optimize the entire feasible schedule.** The multistage schedule is an explicit improvement, not a global optimizer. A rigorous interval or rational branch-and-bound search over arities and cumulative packet exponents could certify a better infimum or a useful restricted optimality statement. Finite thresholds should be included in the objective.

2. **Exploit actual sparse ports.** Full-block Smith profiles are sharp for the full padded orientation. Real packets may contain far fewer ports. A decomposition using those sparse subsets might reduce mode counts without violating source/target locality or making the oracle expensive.

3. **Avoid dense tables when few modes survive.** The new dense algorithm is quadratic. A valuation-aware sparse API could output only requested features or modes below the current precision budget, provided it never discards a unit state that later reappears after a negative-valuation update.

4. **Optimize the time-space tradeoff.** Eight groups and a $3, 3, 2$ block split give a clean proof. Different splits, depth-dependent caching, or bounded-memory recomputation could improve either resource without invoking the unproved global summation oracle.

5. **Close a genuine order-changing interface.** Uniform linear-wire compression, degree-preserving recursive factorization, or a fully costed structured summation method remains the route to a stronger logarithmic denominator. Any claimed solution should include its exact representation and oracle costs.

6. **Formalize the new boundary.** The most contained next formal lemmas are the unit-state recurrence, the missing-bit difference-set lemma, and the compatibility of full-block modes with asymmetric child degrees. They would reduce the trusted gap before attempting a formalized new constant or a complete practical contractor.

Reproduction guide

From the extracted archive, run `python3code/run_all.py`. This uses only Python's standard library and writes fresh results into `reproduced/`. It runs the modular and historical checks and all three resource-certificate checkers. To repeat the timings, run `python3code/benchmark.py--historicalvendor/joint_cauchy.py`. The benchmark prints JSON; redirect it to a new file to preserve the recorded samples. Run `make pdf` to rebuild the report from the bundled LaTeX and figure. Regenerating the scientific figure additionally requires Matplotlib; it is not required to run the verification or build from the included vector PDF. No network access is needed for these steps.

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